

Revised Generation Interconnection System Impact Study Report

For Queue Project Q23-2279

Cwec Iola 138 kV
February 27, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q23-2279. The Interconnection Customer proposes a 118.7 MW battery energy storage facility interconnecting at the Cwec Iola 138 kV substation in the Pennsylvania area. The request was received on January 19, 2023. The study evaluated system intact and single contingency conditions using the regional planning case, including 690 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 6 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$6,303,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Pueblo - Kempster 138kV line	Tuco - John Day Substation 138kV	113.9	235	6.8
Belridge 1B - Pueblo 138kV Line	Tuco - John Day 138kV	112.8	227	6.8
Tuco - Medora 138kV L	Simpson Stuart Tap - Kempster 138kV Line	107.1	339	8.6
John Day - Tuco 138kV	Simpson Stuart Tap - Kempster 138kV	107.0	389	9.9
Matanzas - John Day 138kV	Simpson Stuart Tap - Kempster 138kV	102.5	396	9.9
Kempster - Simpson Stuart Tap 138kV	Tuco Substation - John Day 138kV	101.4	499	9.6

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Pueblo - Kempster 138kV	Rebuild line with higher-capacity conductor	\$2,220,000	100% (\$2,220,000)
Pueblo - Belridge 1B 138kV L	Rebuild line with higher-capacity conductor	\$3,352,000	100% (\$3,352,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Tuco - Medora 138kV	Reconductor line	\$467,000	20% (\$94,000)
John Day - Tuco 138kV	Reconductor line	\$753,000	16% (\$123,000)
Matanzas - John Day 138kV	Reconductor line	\$958,000	16% (\$157,000)
Kempster - Simpson Stuart Tap 138kV	Reconductor line	\$1,810,000	20% (\$357,000)
	Total allocated to project		\$6,303,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.